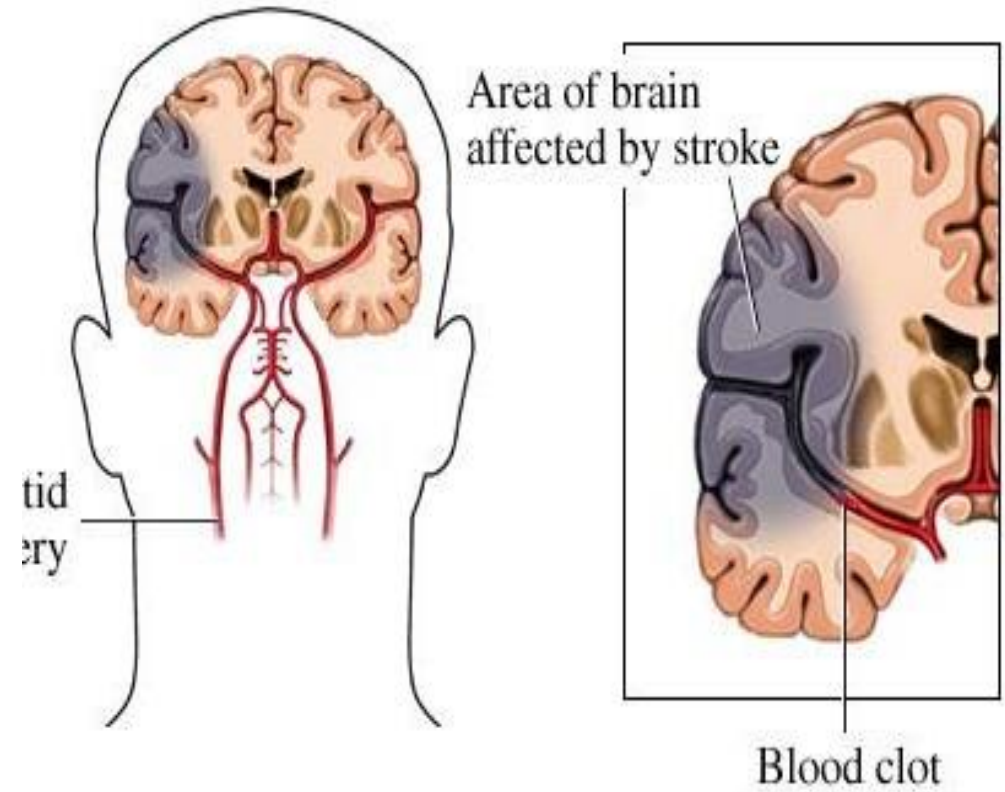


Stroke



Prepared by
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Stroke

- ***Outline***
- Classification of Stroke:
- Etiology of Stroke
- Risk factors for stroke
- Clinical Manifestations of stroke
- Diagnosis of Stroke
- Management of Stroke
- Nursing Management of Stroke

Pre test

Mention causes
of stroke

Mention types
of stroke

Stroke

- Stroke is the sudden onset of acute neurological deficit persisting for more than 24 hours and caused by the interruption of blood flow to a region of the brain.
- Blood flow is disrupted because of an obstruction of a vessel, a thrombus or embolus, or the rupture of a vessel.
- The apparent clinical features depend on the location of the event and region of the brain perfused by the vessel.

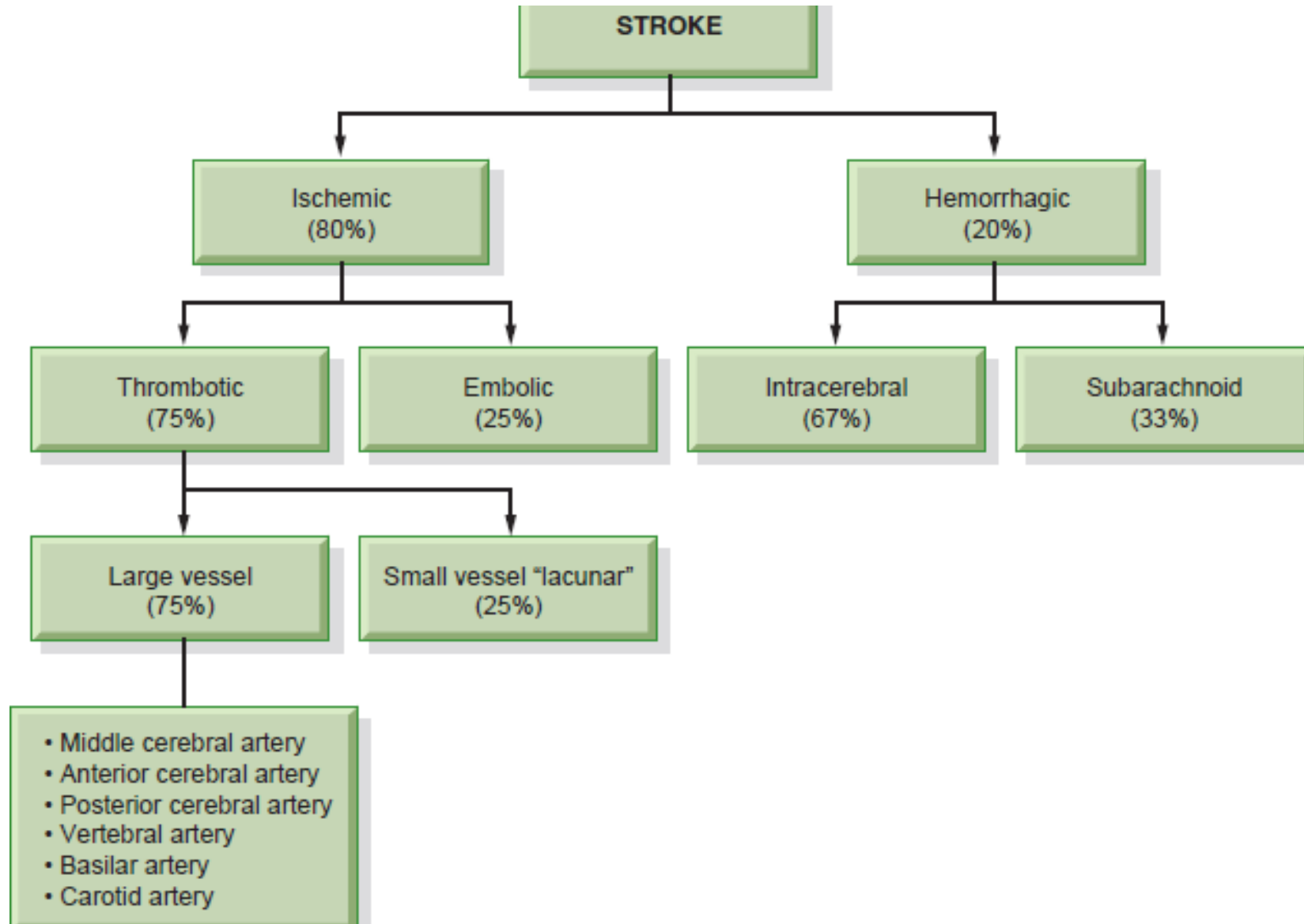


Classification of Stroke:

- **Ischemic Stroke** (87% of all strokes)
 - **Hemorrhagic** stroke include:
 - Subarachnoid hemorrhages (3% SAHs)
 - Intracerebral hemorrhages (10%)
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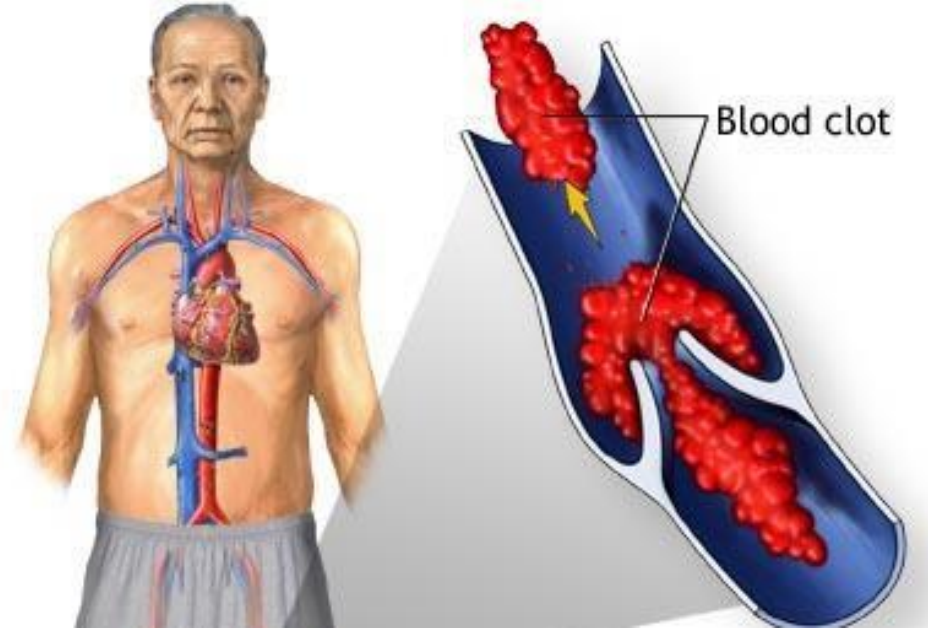
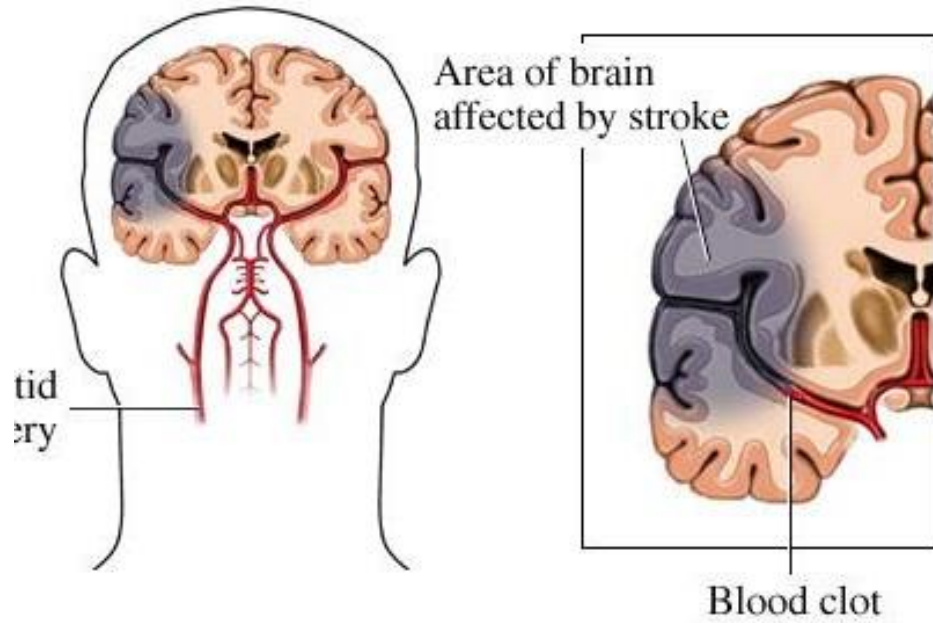
Table 62-1 • Major Types of Stroke and Their Causes

CLASSIFICATION	CAUSES
Ischemic	Large artery thrombosis Small penetrating artery thrombosis Cardiogenic embolic Cryptogenic (no known cause) Other
Hemorrhagic	Intracerebral hemorrhage Subarachnoid hemorrhage Cerebral aneurysm Arteriovenous malformation



Etiology of Stroke

- **Ischemic strokes** are due to vascular obstruction (thrombi or emboli), resulting in ischemia and infarction.
- **Hemorrhagic Stroke are due to:**
 - Hypertensive vascular disease (which causes an intracerebral hemorrhage),
 - **A ruptured aneurysm** is a dilation of the walls of a cerebral artery that develops as a result of weakness in the arterial wall.
 - **A ruptured an Arteriovenous Malformation (AVM)** due to an abnormality in embryonal development that leads to a tangle of arteries and veins in the brain without a capillary bed.
 - The absence of a capillary bed leads to dilation of the arteries and veins and eventual rupture.
 - They are commonly a cause of hemorrhage in young people.



Types of shock

Risk factors for stroke

- Risk factors include:
 - smoking,
 - hypertension,
 - obesity,
 - cardiac disease,
 - hypercholesterolemia,
 - diabetes,
 - cancer,



Clinical Manifestations of stroke

- Manifestations depend on the anatomical location of the lesion; an infarct in a certain portion of the brain results in loss of function of the body part.
 - If symptoms resolve in less than 24 hours, the event is classified as a ***Transient Ischemic Attack (TIA)***,
 - is a deficit lasting less than 24 hours that is attributed to focal cerebral or retinal ischemia and last for only minutes to less than an hour.
-

Manifestations of Ischemic Stroke

- **Early Sign**
- Numbness or weakness of the face, arm, or leg, especially on one side of the body
- Confusion or change in mental status
- Trouble speaking or understanding speech
- Visual disturbances
- Difficulty walking, dizziness, or loss of balance or coordination
- Sudden severe headache

Visual Field Deficits

- **Homonymous hemianopsia:**
 - (loss of half of the visual field) Unaware of persons or objects on side of visual loss
 - Neglect of one side of the body Difficulty judging distances
- **Loss of peripheral vision:**
 - Difficulty seeing at night Unaware of objects or the borders of objects
- **Diplopia:** Double vision.

Motor Deficits

- **Hemiparesis:** Weakness of the face, arm, and leg on the same side (due to a lesion in the opposite hemisphere)
 - **Hemiplegia:** Paralysis of the face, arm, and leg on the same side (due to a lesion in the opposite hemisphere)
 - **Ataxia:**
 - Staggering, unsteady gait Unable to keep feet together; needs a broad base to stand
 - **Dysarthria:** Difficulty in forming words
 - **Dysphagia:** Difficulty in
 - swallowing
-



Sensory and verbal Deficits

- **Paresthesia** (occurs on the side opposite the lesion)
 - Numbness and tingling of extremity
 - Difficulty with proprioception
 - **Verbal Deficits**
 - **Expressive aphasia:** Unable to form words that are understandable; may be able to speak in single-word responses
 - **Receptive aphasia:** Unable to comprehend the spoken word; can speak but may not make sense
 - **Global (mixed) aphasia**
 - :Combination of both receptive and expressive aphasia.
-

Cognitive and Emotional deficit

- **Cognitive Deficits**

- Decreased attention span Impaired ability to concentrate Poor abstract reasoning
- Altered judgment

- **Emotional Deficits**

- Loss of self-control Emotional lability
- Decreased tolerance to stressful situations
- Depression Withdrawal
- Fear, hostility, and anger Feelings of isolation

Manifestations of Hemorrhagic Stroke

Early signe

“Exploding headache”

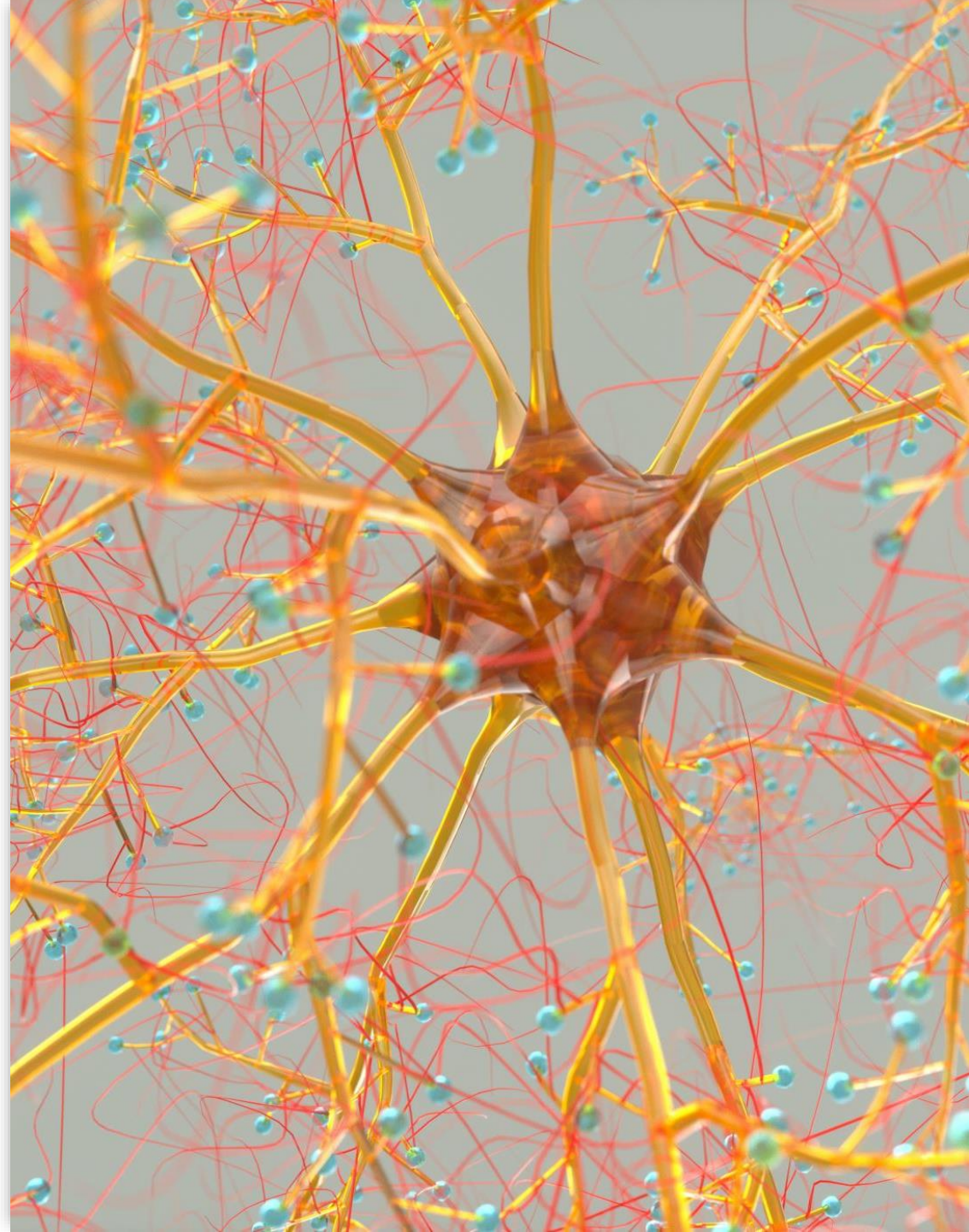
Decreased level of
consciousness

Visual and motor deficit

- The patient with a hemorrhagic stroke can present with a similar neurologic deficits of ischemic stroke (the same motor, sensory, cranial nerve, cognitive)
- The conscious patient most commonly reports a severe headache
- Patients with acute **intracerebral hemorrhage** have:
- Vomiting
- An early sudden change in level of consciousness,
- Focal seizures due to frequent brain stem involvement
- Patients with an **intracranial aneurysm or AVM** may have some unique clinical manifestations.
- A sudden, unusually severe headache and often loss of consciousness for a variable period of time.

Sensory deficit

- There may be pain and rigidity of the back of the neck (nuchal rigidity) and spine due to meningeal irritation.
- Visual disturbances (visual loss, diplopia, ptosis) occur if the aneurysm is adjacent to the oculomotor nerve.
- Tinnitus, dizziness, and hemiparesis may also occur.
- In other cases, severe bleeding resulting cerebral damage, followed rapidly by coma and death

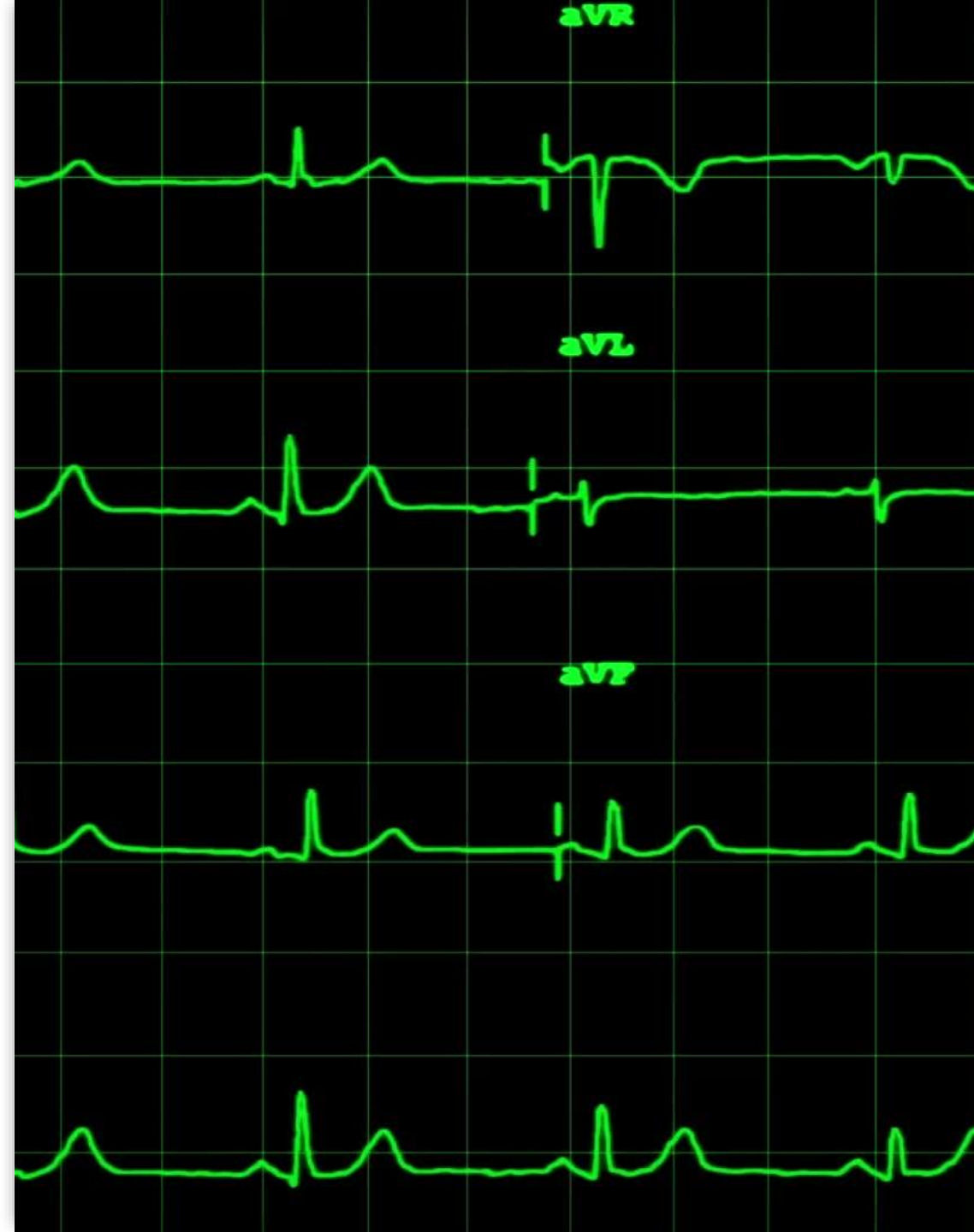


Diagnosis of Stroke

- **Rapid diagnosis of a stroke** is essential so that appropriate patients can receive thrombolytic therapy, the goal of which is to save damaged brain tissue and minimize permanent deficits.
- **Diagnostic test of stroke includes:**
- **CT** scanning without contrast media is to differentiate intracranial hemorrhage from ischemic stroke.
- **MRI** can demonstrate infarction of cerebral tissue earlier than CT and MRI determine the type of stroke, the size and location of the hematoma, and the presence or absence of ventricular blood and hydrocephalus.

- **CT angiography** or magnetic resonance imaging and angiography (MRI) of the brain and neck vessels confirm the diagnosis of an intracranial aneurysm or AVM.
- These tests show the location and size of the lesion and provide information about the affected arteries, veins, adjoining vessels, and vascular branches.
- Other tests includes:
- **transcranial Doppler flow studies**; transthoracic or transesophageal echocardiography; xenon-enhanced CT scan; and Single Photon Emission CT (SPE CT) scan.

- 12-lead electrocardiography, chest radiography, and continuous cardiac monitoring are suggested to detect a cardiac cause or coexisting condition.
- Echocardiography is valuable in identifying a cardioembolic phenomenon.
- Laboratory evaluation of hematological function, electrolyte and glucose levels, and renal and hepatic function is also recommended.
- Arterial blood gas analysis is performed if hypoxia is suspected.
- An electro-encephalogram is obtained if seizures are suspected.



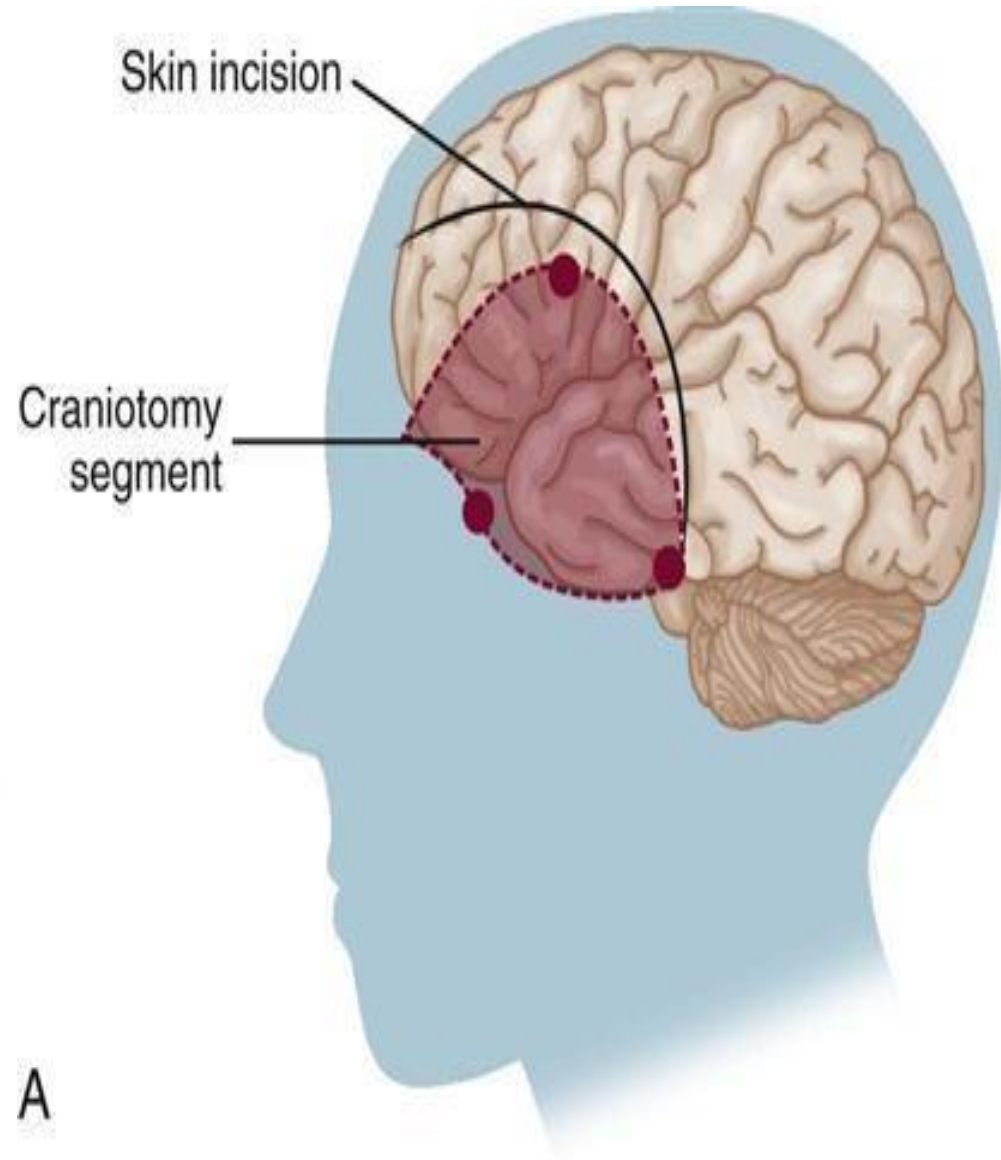
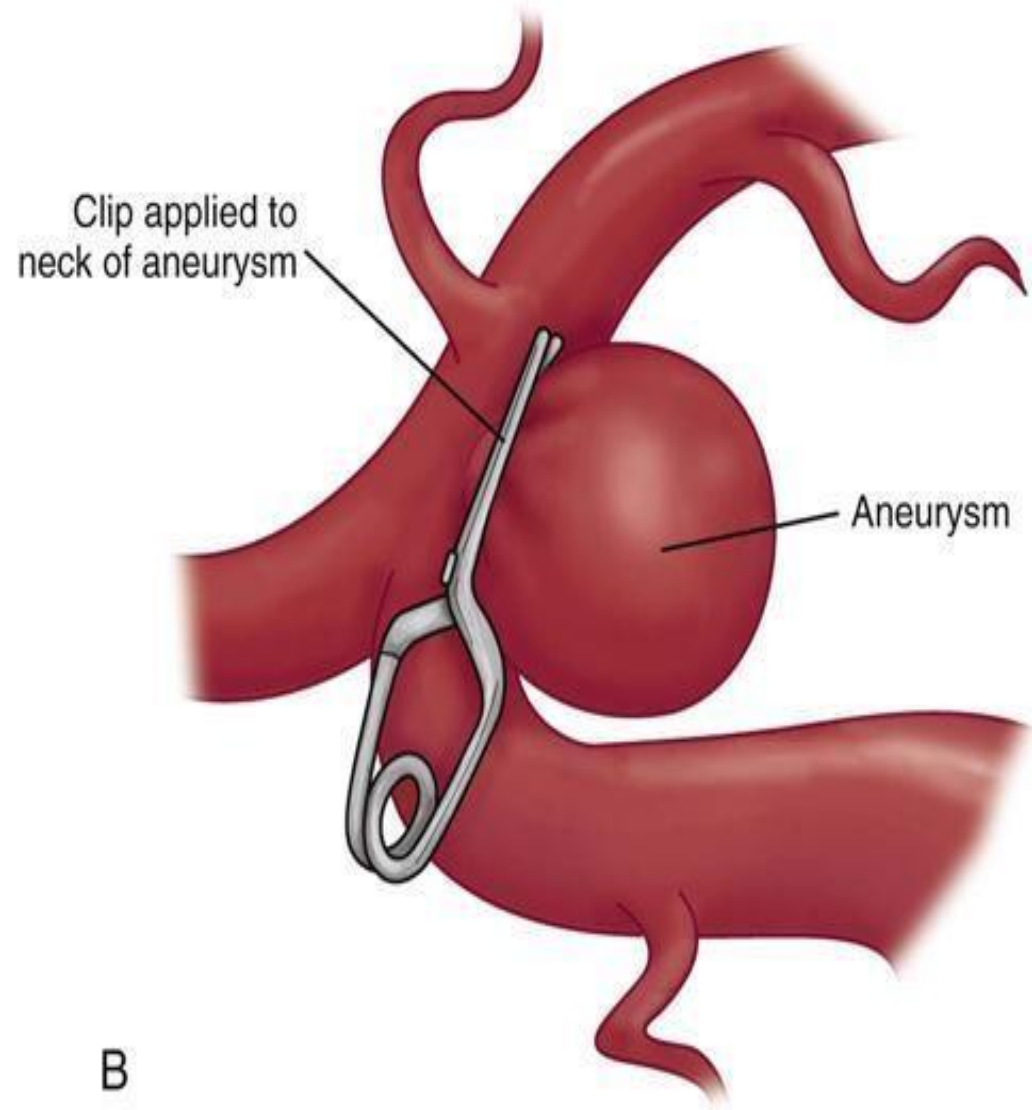
Management of Stroke

- **Treatment of Ischemic Stroke**
- **Medical management** to restore of Cerebral Blood Flow, maintain reperfusion and Prevent recurrent of thrombosis.
- **Medical treatment includes:**
- thrombolytic therapy with intravenous recombinant tissue- type plasminogen activator (rtPA), anticoagulant ((IV heparin or low- molecular-weight heparin) and antiplatelet drugs (Plavix, Aspirin).
- **Surgical prevention** performing Carotid Endarterectomy surgery

Treatment of Hemorrhagic Stroke

- **Surgical Management**

- to allow the brain to recover from the initial insult (bleeding), to prevent or minimize the risk of re-bleeding, and to prevent or treat complications.
- **The surgical procedure involves:**
 - A **craniotomy** to evacuate bleeding or hematoma and expose or isolate the area of aneurysm and Arteriovenous Malformations (AVM).
- **Surgical *clipping* or endovascular coiling** with complete obliteration of the aneurysm.



Medical Management of Ischemic Stroke

Thrombolytic Therapy

- **Action of thrombolytic medications** is exogenous drugs that dissolve clots.
- The t-PA activates plasminogen, a naturally enzyme present in the intravascular endothelium that protects against excessive clotting.
- Activating plasminogen initiates the process of dissolving the clot through fibrinolysis.
- The maximal time is now expanded from 3 to 4.5 hours or less from the onset of the neurological symptoms.

Dose and Route:

- The recommended dose of t-PA is 0.9 mg/kg up to a maximum dose of 90 mg.
- The 10% of the total dose as a bolus over the total dose is administered as an initial intravenous bolus over one to two minutes, and the remaining 90% is administered by intravenous infusion over 60 minutes.
- No other antithrombotic therapy should be given for the next 24 hours.
- A major risk of this therapy is intracerebral hemorrhage.

Inclusion Criteria of thrombolytic therapy

- Clinical diagnosis of ischemic stroke
- Time of onset of stroke known and is 3 hours or less
- Systolic blood pressure ≤ 185 mm Hg; diastolic ≤ 110 mm Hg
- Not a minor stroke or rapidly resolving stroke
- No seizure at the onset of stroke
- Not taking warfarin (Coumadin)
- Prothrombin time ≤ 15 seconds or INR ≤ 1.7

Inclusion Criteria of thrombolytic therapy

- Not receiving heparin during the past 48 hours with elevated partial thromboplastin time
- Platelet count $\geq 100.000/\text{mm}^3$
- No prior intracranial hemorrhage, neoplasm, arteriovenous malformation, or aneurysm
- No major surgical procedures within 14 days
- No stroke, serious head injury, or intracranial surgery within 3 months
- No gastrointestinal or urinary bleeding within 21 days

- **Side effects:**

- Bleeding and hypotension is the most common side effect of t-PA administration.
- Intracranial bleeding is a major complication that occurred in approximately 6.4% of patients.

- **Nursing Consideration**

- Vital sign are obtained every 15 minutes for the first 2 hours, every 30 minutes for the next 6 hours, then every hour for 16 hours.
- Measure blood pressure, systolic should be less than 180 mm Hg and diastolic less than 100 mm Hg.
- Measure arterial blood gases



The patient is closely monitored for any bleeding (IV insertion sites, urinary catheter site, endotracheal tube, nasogastric tube, urine, stool, emesis, other secretions).

A 24-hour delay in placement of nasogastric tubes, urinary catheters, and intra-arterial pressure catheters is recommended.

Anti- Coagulant

- Secondary treatment options for stroke include anticoagulation with antithrombotic and anti-platelet drugs.
- If a patient experiences atrial fibrillation, anticoagulation with warfarin may be warranted. The patient will need instruction about bleeding precautions.

- Education about the purpose of the medication, information about moderate consumption of leafy green vegetables containing vitamin K.
- The importance of having blood drawn regularly to monitor prothrombin time and the INR.
- For safety patients should be instructed to obtain medical identification cards and bracelets that can be identified as taking an anticoagulant in the event of a medical emergency.

Antiplatelet drugs

- These drugs prevent platelets from adhering to the wall of an injured blood vessel or other platelets and to prevent a future thrombotic or embolic event.
- **Dipyridamole** increases the effect of specific factors that act as anti aggregates to reduce platelet aggregation.
- **Ticlopidine** inhibits platelet function by suppressing adenosine diphosphate induced platelet aggregation and aggregation due to other factors.
- The recommended dose of ticlopidine is 250 mg twice a day. Neutropenia and
- thrombocytopenia are known serious adverse effects, so it is now rarely used.



- **Clonidogrel**

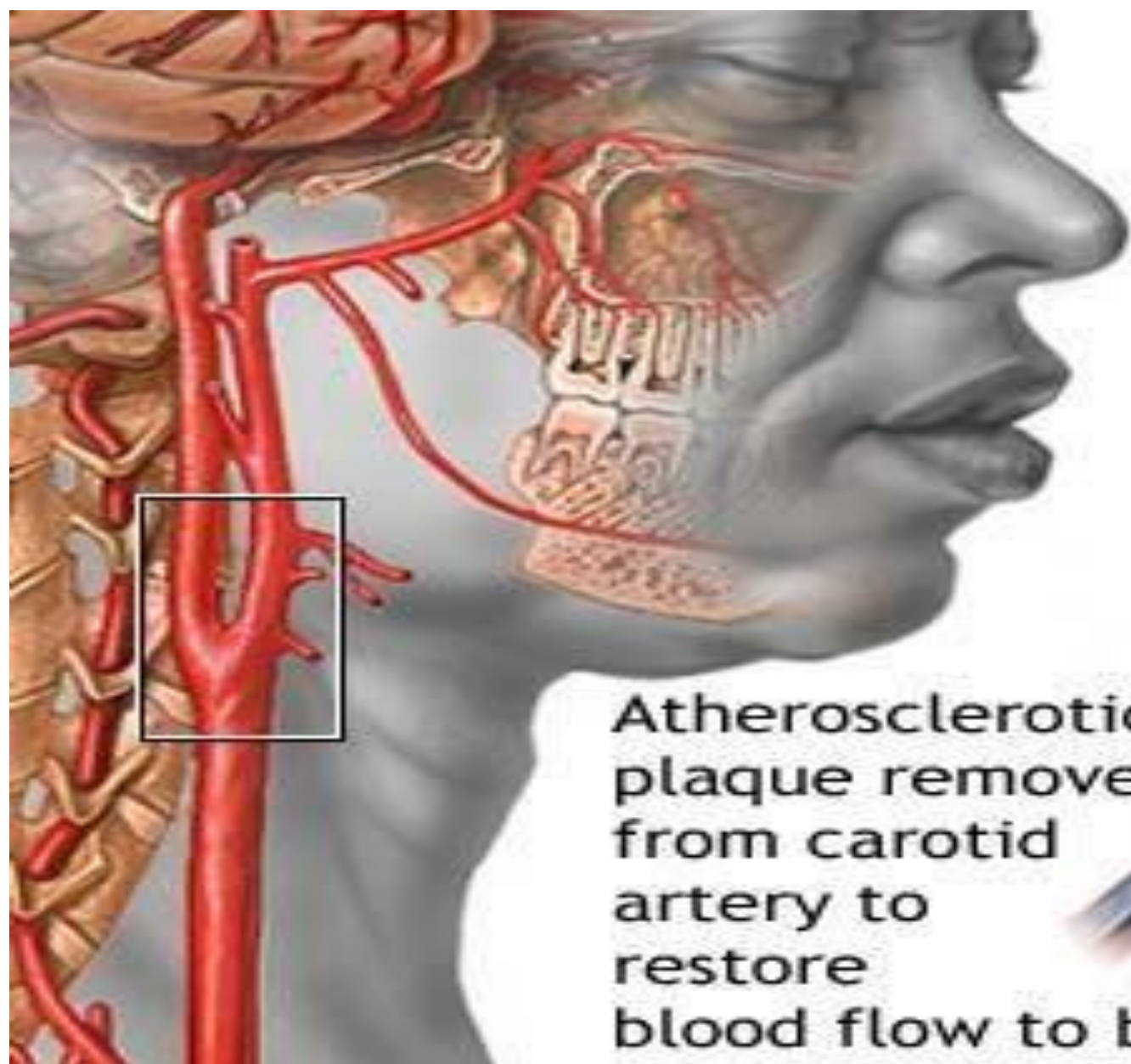
- also inhibits the activity of adenosine diphosphate but is not associated with an increased risk for neutropenia.

- **Aspirin**

- limits platelet adhesion and aggregation.
- The suggested dose of aspirin is 81 to 325 mg/d.

Surgical Management of Ischemic Stroke

- *Carotid endarterectomy* Surgery
 - Is the removal of an atherosclerotic plaque or thrombus from the carotid artery to prevent stroke in patients with occlusive disease of the extracranial cerebral arteries.
 - This surgery is indicated for patients with symptoms of TIA or mild stroke found to be caused by severe (70% to 99%) carotid artery stenosis or moderate (50% to 69%) stenosis with other significant risk factors.



Atherosclerotic
plaque removed
from carotid
artery to
restore
blood flow to brain



Complications Post Endarterectomy

- **Incision hematoma:**

- Occurs in 5.5% of patients. Large or rapidly expanding hematomas require emergency treatment.
- If the airway is obstructed by the hematoma, the incision may be opened at the bedside.

- **Hypertension:**

- Poorly controlled hypertension increases the risk of postoperative complications, including hematoma and hyperperfusion.

- syndrome. There is an increased incidence of neurologic impairment and death due to intracerebral hemorrhage.
- **Postoperative hypotension:**
- Occurs in approximately 5% of patients, treated with fluids and low-dose phenylephrine infusion.
- Usually resolves in 24–48 h. Patients with hypotension should have serial ECGs to rule out myocardial infarction.

Hyperperfusion Syndrome:

- Occurs when cerebral vessel autoregulation fails.
- Arteries adapted to diminished blood flow may be permanently dilated; increased blood flow after endarterectomy attached with insufficient vasoconstriction leads to capillary bed damage, edema, and hemorrhage.

- **Intracerebral hemorrhage:**

- Occurs infrequently, but is often fatal (60%) or results in serious neurologic impairment.
- It can occur secondary to hyperperfusion syndrome.
- Increased risk with advanced age, hypertension, presence of high-grade stenosis, poor collateral flow, and slow flow in the region of the middle cerebral artery.

Nursing Management Post Endarterectomy

- The primary complications of carotid endarterectomy are stroke, cranial nerve injuries, infection or hematoma at the incision, and carotid artery disruption.
- * It is important to maintain adequate blood pressure levels in the immediate postoperative period.
- *Hypotension is avoided to prevent cerebral ischemia and thrombosis.
- *Uncontrolled hypertension may precipitate cerebral hemorrhage, edema, hemorrhage at the surgical incision, or disruption of the arterial reconstruction.
- Medications are used to reduce the blood pressure to

- *Close cardiac monitoring is necessary, because these patients have a high incidence of coronary artery disease.
- *The surgeon is notified immediately if a neurologic deficit develops.
- *Formation of a thrombus at the site of the endarterectomy is suspected if there is a sudden increase in neurologic deficits, such as weakness on one side of the body. The patient should be prepared for repeat endarterectomy.
- *Difficulty in swallowing, hoarseness or other signs of cranial nerve dysfunction must be assessed.
- The nurse focuses on assessment of the following cranial nerves: Facial (VII), Vagus (X), spinal accessory (XI), and Hypoglossal (XII).
- *Emergency airway supplies, including those needed for a tracheostomy must be available

Nursing Management of Stroke

- **Monitoring for Changes in Neurological and Hemodynamic Status** The goal of frequent assessments is early recognition of neurological or hemodynamic deterioration.
- Close monitoring of the patient's neurological signs and vital signs is essential and requires almost continuous observation.
- Automatic noninvasive devices such as a blood pressure cuff and a pulse oximeter are helpful.
- Seizure activity must be identified and treated immediately and the physician be notified at the first sign of any changes.

- **Maintaining Surveillance for Complications**

- The patient with stroke should be monitored closely for signs of bleeding, vasospasm, and increased ICP.
 - Other complications of stroke include aspiration, malnutrition, pneumonia, DVT, pulmonary embolism, pressure ulcers, contractures, and joint abnormalities.
- 

Increased Intracranial Pressure:

- Numerous signs and symptoms of increased ICP can be observed.
- A change in the level of consciousness is the most sensitive indicator.
- Others include unequal pupil size, decreased pupillary response to light, headache, projectile vomiting, altered breathing patterns, Cushing's triad (bradycardia, systolic hypertension, and bradypnea), diminished brainstem reflexes, papilledema, and abnormal extension (decerebrate posturing) or flexion (decorticate posturing).

Impaired Swallowing

- : Normal swallowing occurs in four phases that are controlled by the cranial nerves.
- Damage to the brain, brainstem, or cranial nerves can result in a variety of swallowing deficits that can place the patient at risk for aspiration.
- The stroke patient is observed for signs of dysphagia, including drooling; difficulty handling oral secretions; absence of gag, cough, or swallowing reflex; moist, gurgling voice quality; decreased mouth and tongue movements; and the presence of dysarthria.
- A speech therapy consultation is initiated if any of these signs is present, and the patient must not be orally fed.
- In the absence of these warning signs, the patient may be fed, as ordered by the physician, although he or she must be continually monitored for signs of aspiration.

Educating the Patient and Family

- Rehabilitation starts in the critical care area, with a multi-disciplinary team designing and implementing an individualized plan to maximize the patient's potential for neurological rehabilitation.
- Early in the patient's hospital stay, the patient and family must be taught about stroke, its causes, and its treatment.
- As the patient moves toward discharge, teaching focuses on the interventions necessary for preventing the recurrence of the event and on maximizing the patient's rehabilitation potential.
- The patient's family must be encouraged to participate in the patient's care; learn how to feed, dress, and bathe the patient; and learn some basic rehabilitation techniques.
- The importance of participating in a neurological rehabilitation program or a support group, or both, must be stressed.

Principles of rehabilitation after hemorrhagic stroke

- To **reduce disability**
- **Multidisciplinary team care** : medical, nursing, physical and occupational therapy, speech and language pathology, and social work staff
- **Comprehensive assessment** of motor, cognitive, and communication functions, swallowing, and emotions

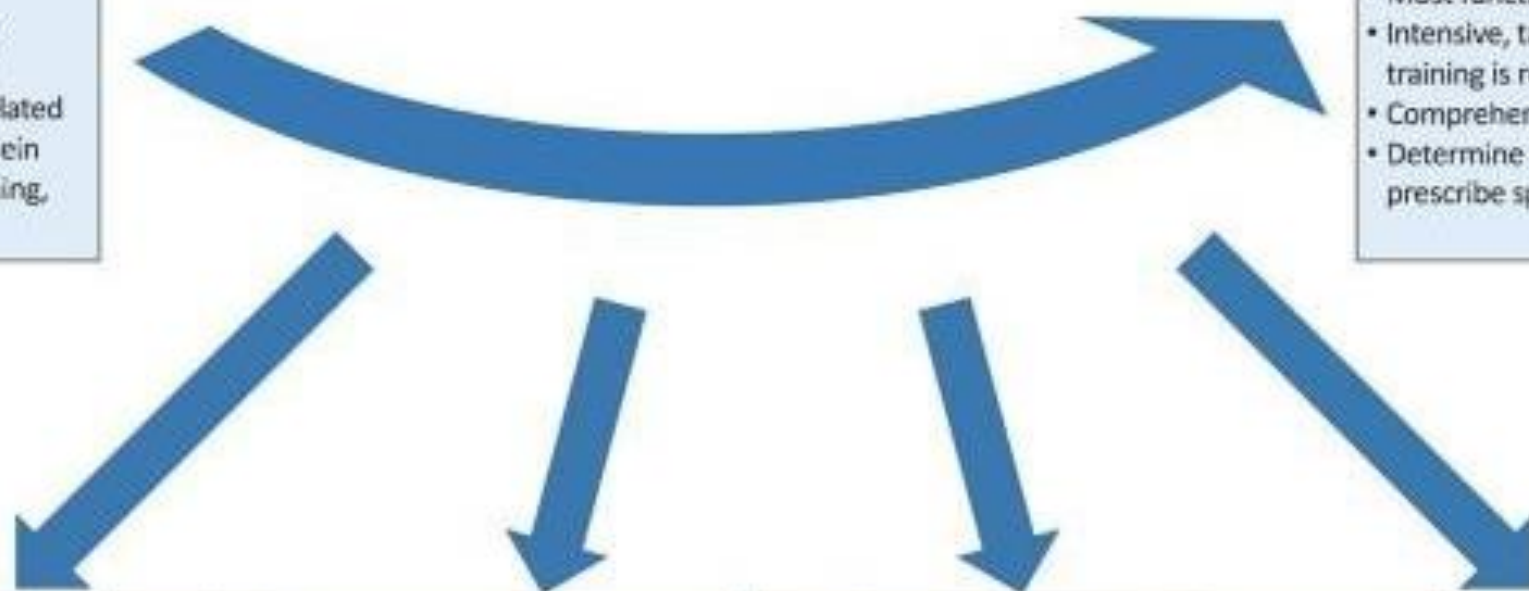
Acute stage

Early mobilization

- Changing positions, sitting up in bed, transferring to a wheelchair, walking
- Preventing or reducing immobility-related complications (e.g. infections, deep vein thromboembolism, falls, deconditioning, muscle atrophy)

Subacute to chronic stage

- Most functional recovery takes place
- Intensive, task-specific, and repetitive training is required
- Comprehensive stroke rehabilitation unit
- Determine the goals of rehabilitation and prescribe specific treatment



Motor impairment

- Proper positioning
- Passive range of motion exercise
- Neurodevelopmental (NDT) exercise
- Rood technique
- Proprioceptive neuromuscular facilitation (PNF)
- Constraint-induced movement therapy (CIMT)
- Neuromuscular electrical stimulation (NMES)

Dysphagia

- Compensatory strategy: nasogastric tube, chin tuck maneuver, head turn maneuver, Mandelsohn's maneuver
- Rehabilitative strategy: oral sensory stimulation, transcutaneous neuromuscular electrical stimulation (NMES), shaker exercise

Speech and language disorders

- Non-fluent aphasia: melodic intonation therapy
- Constraint induced aphasia therapy (CIAT)
- Dysarthria/dysphonia: Respiratory training, strengthening oromotor speech muscles, sensory stimulation

Cognitive impairment

- Individualized program
- Pharmacologic interventions: AchE inhibitors, NMDAR inhibitors

Patient Education Plan

Before the patient's discharge, the patient should be able to teach back the following topics:

Pathophysiology of the disease

Specific cause

Risk factor modification

Importance of taking medications

Activities of daily living

Measures to prevent injuries of impaired limbs

Measures to compensate for residual deficits

Basic rehabilitation techniques

Importance of participating in neurologic rehabilitation program or support group

Examples of Nursing Diagnosis

- Risk for Ineffective Cerebral Tissue Perfusion.

Acute Pain related to transmission and perception of cutaneous, visceral muscular, or ischemic impulses.

Unilateral Neglect related to perceptual disruption

Impaired Verbal Communication related to cerebral speech center injury

Impaired Swallowing related to neuromuscular impairment, fatigue, and limited awareness.

Risk for Aspiration.

Risk for Infection.

Anxiety related to threat to biologic, psychologic, or social integrity.

Disturbed Body Image related to actual change in body structure, function, or appearance.

Compromised Family Coping related to critically ill family member.

Post test

Mention
types of
stroke

Mention
complication
of stroke



Any question



Thank you